

NZGDC 2026

Designing Shared Worlds Across Distance

What Multiplayer AR Research Taught Me

Yasas Sri Wickramasinghe
HIT Lab NZ · University of Canterbury





INTRO



SHOW



PLAY



RULES



BUILD



WRAP-UP



Q&A

K I A O R A

Yasas Sri Wickramasinghe

PhD, Human Interface Technology · University of Canterbury

Research it.

Postdoctoral researcher, HIT Lab NZ
XR research with Sony Interactive Entertainment

Build it.

Co-founded NavitaZ VR Labs
Niantic Studio developer accelerator grant, 2024

Teach it.

Senior Lecturer, Yoobee College
Previously University of Moratuwa, Sri Lanka

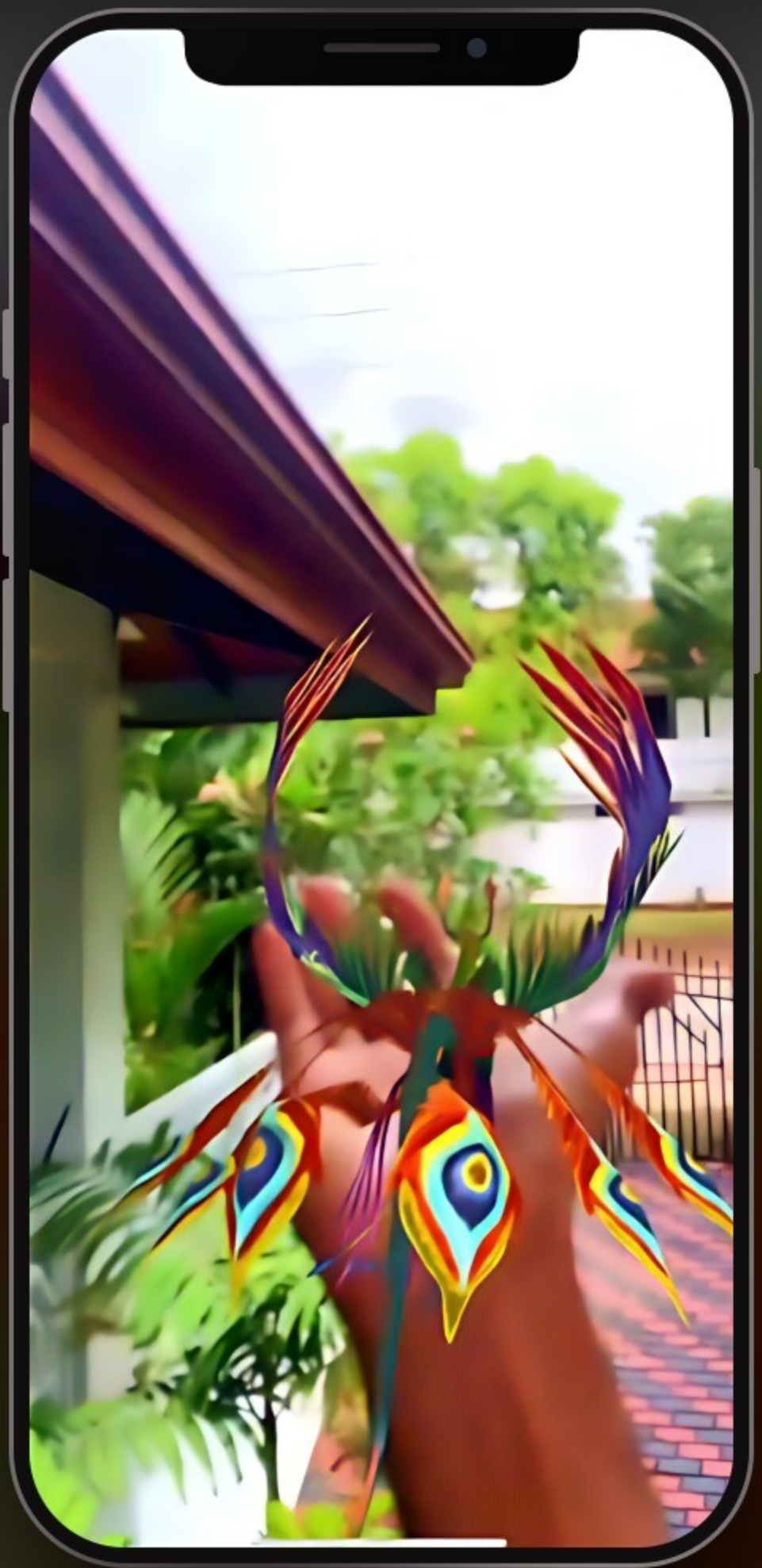


Everything from this talk

Slides, papers, demos & the six rules

yasassri.me/nzgdc

www.yasassri.me · linkedin.com/in/yasassri

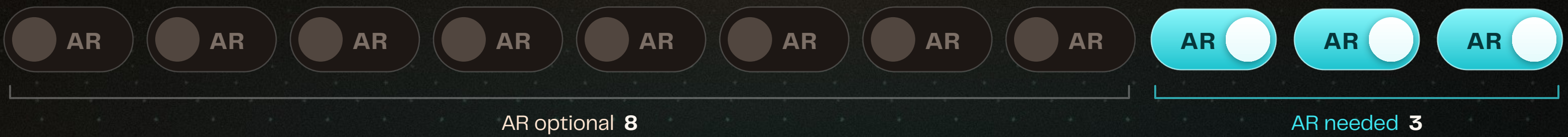


QUICK GUESS

Researchers studied 11 popular AR games.

In how many was the AR optional?

8 of 11
had AR the game didn't need



Removing the AR “would make the game more streamlined and less laborious.”

Source: Alha, Leorke, Koskinen & Paavilainen (2023), Augmented play, Convergence 29(2), 342-362, p. 355. The 11 were popular location-based games with AR features (1M+ downloads, a 4-star+ rating or \$5,000+ revenue, Feb 2021).

The problem - part 1

Most players turn the AR camera off

~7%

of surveyed location-based game players regularly used the AR camera

Main reason

“As it is slower to use, I prefer not to.”

AR slowed players down, so they switched it off.

Across 11 location-based games with AR features: “AR is most commonly a gimmick rather than a central element of the game experience.”

Alha, Leorke, Koskinen & Paavilainen (2023), Convergence



What if the game didn't work without AR?

The problem - part 2

People want to play together remotely

2020

Pokémon GO added Remote Raid Passes, so players could join raids from home.

US\$1B+

Pokémon GO player spending in the first ten months of 2020, its best year to that point.

But remote play felt like a lobby, not a place.

You could join a raid at a location, but you weren't really there with your friends.

RECENT RESEARCH

Xu et al., CHI 2023

Interviews with 41 Pokémon GO players: raid groups relied on outside social media groups to coordinate local and remote players.

Can AR help two players in different places feel they share one space?



WINDOW



OVERLAY

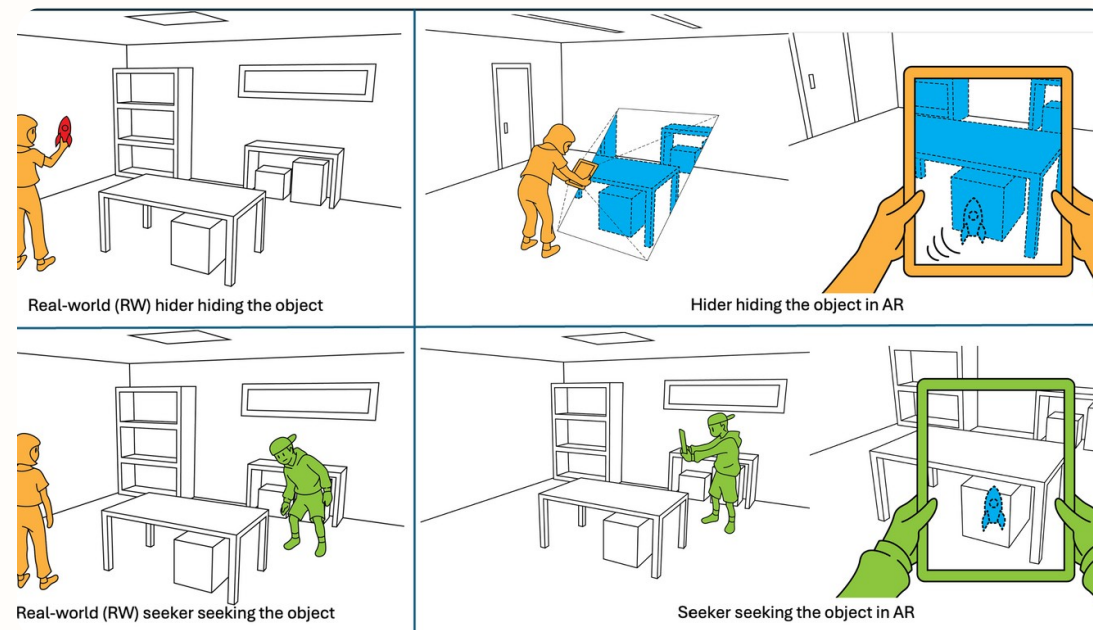


TABLETOP

Game 1 · Remote resource sharing

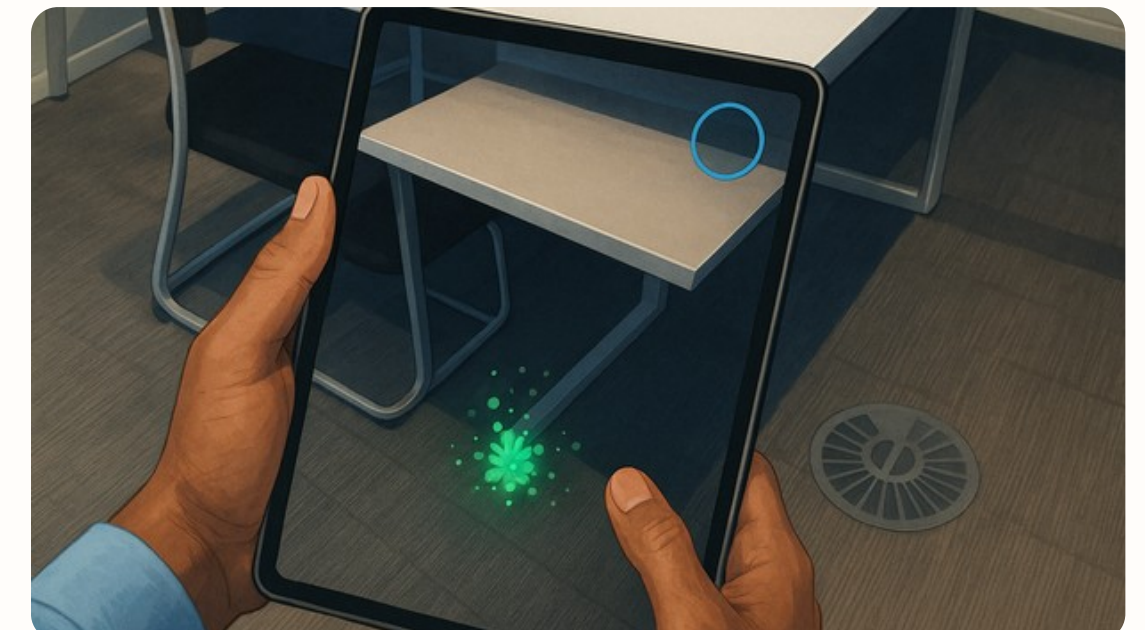
30 players · three ways to show a remote room

Entertainment Computing, 2025



Game 2 · Remote AR hide-and-seek

60 players · AR vs real-world hide-and-seek
IEEE VR 2025 · MTI journal, 2025



Game 3 · Mechanics only AR can do

38 players · resize, clone and swap
OzCHI 2025

Visualizing Remote Spaces in AR

Show me your world

Before a friend can play in your space, you have to decide how they see it. That turned out to be a game design decision, not only a rendering one.



Game 1

Three Different Representations for Remote Space Representations in AR



WINDOW



OVERLAY



TABLETOP



A player collecting real-world resources outdoors

Game 1 - what we found

Life-size views felt like being there

Which view did players prefer? (n = 30)

Overlay 20 67%

Window 9 30%

Tabletop 1 3%

8 of 30

players tried to walk through the Window into the remote room, and some called it "the portal" without being told.

"The portal feels like my friend's space is also part of my real world."

Participant 6

Overlay was the only view that significantly raised "self-location", the feeling of being in the other place (compared with Tabletop).
Only one player preferred Tabletop.

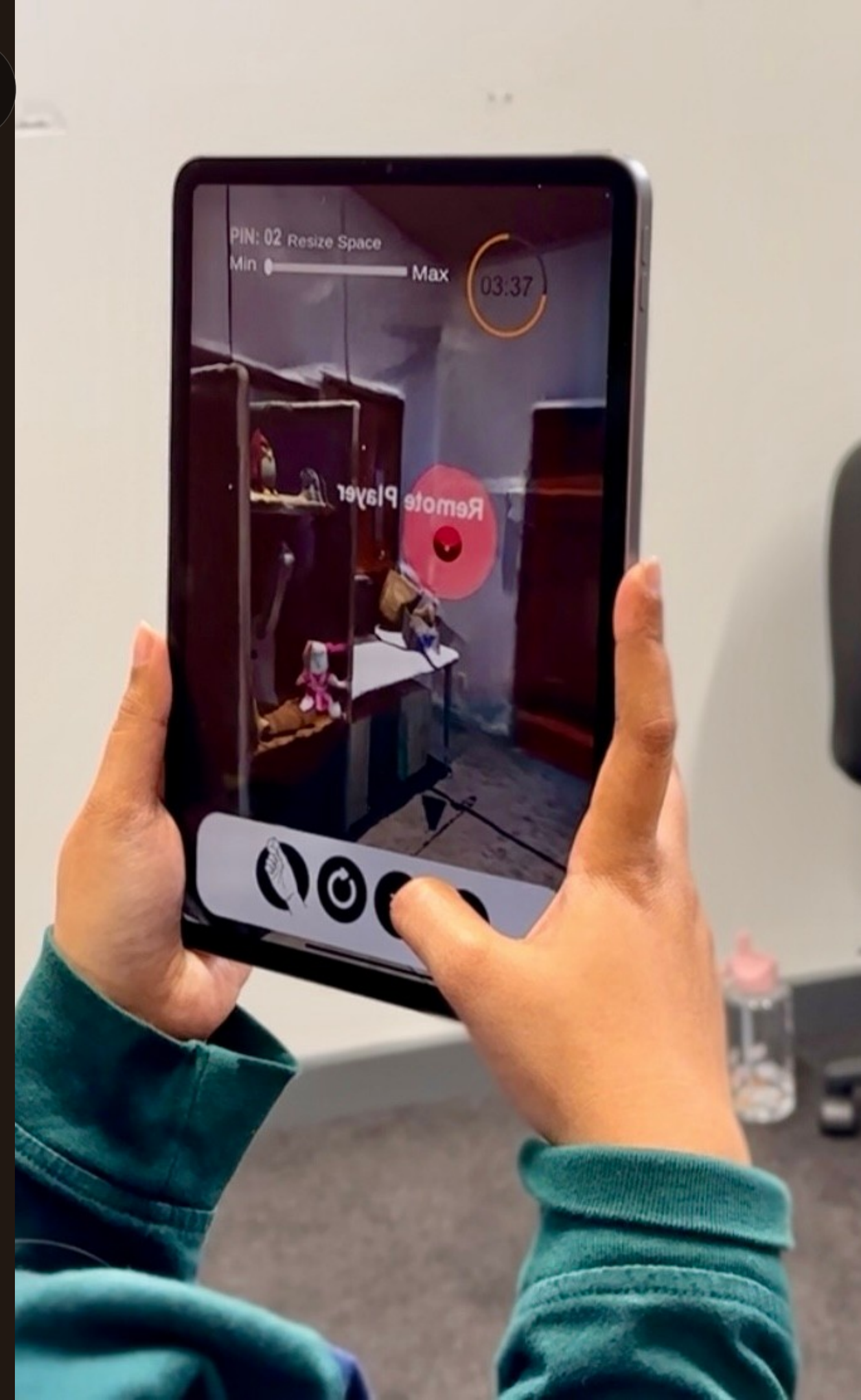
RULE

Life-size when you want players to feel there. A window when they need to get things done.

CHAPTER 2

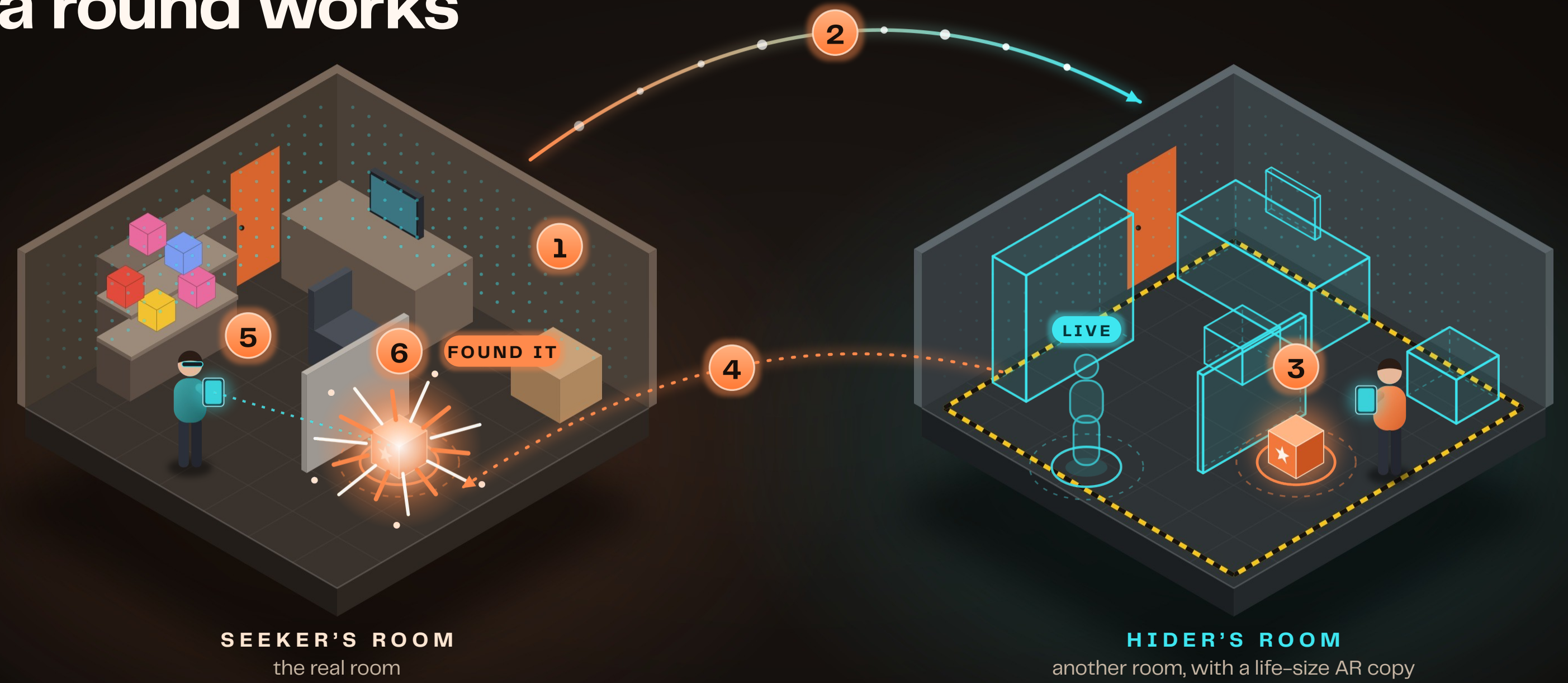
Hide-and-seek across distance

Showing a place is one step. Next, we played a game in it: hide-and-see, which everyone already knows.



GAME 2 · REMOTE AR HIDE-AND-SEEK

How a round works



1 SCAN

The seeker's room is scanned with a LiDAR iPad.

2 COPY

The hider sees that room life-size, in their own room.

3 HIDE

The hider hides a virtual toy in the copy.

4 SYNC

The toy appears in the same spot in the real room.

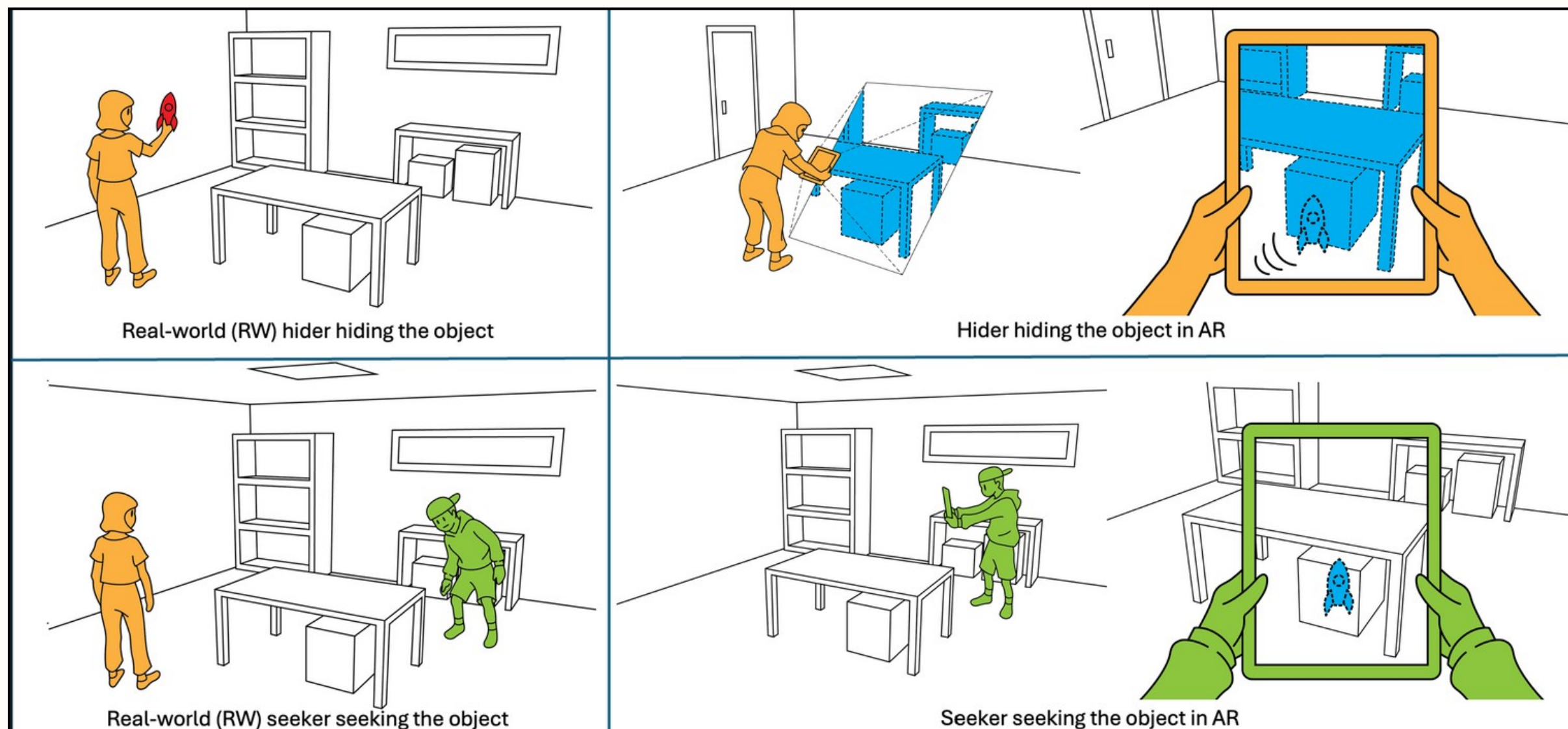
5 SEARCH

The seeker searches with an iPad. The hider watches live.

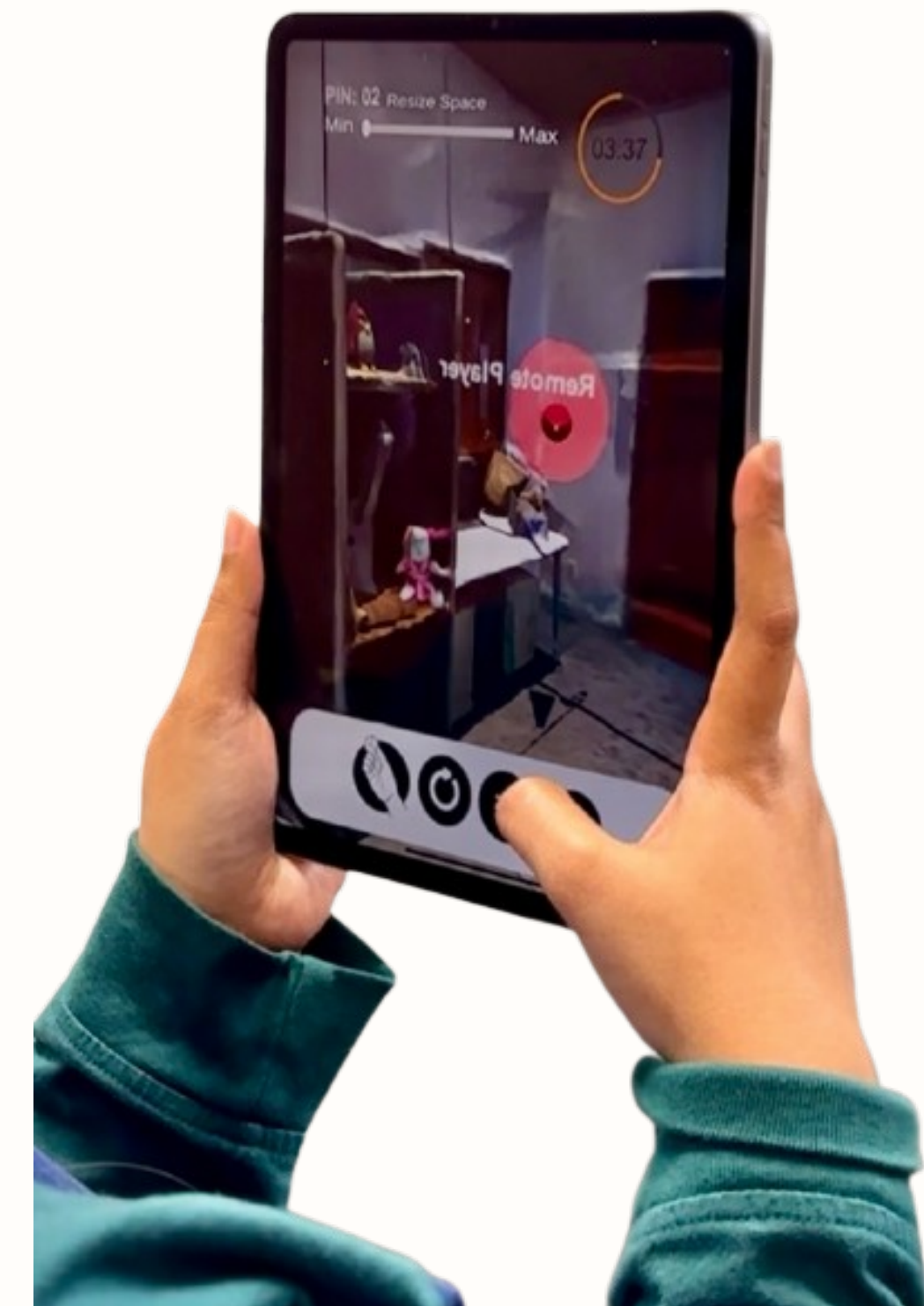
6 FOUND

Seeker finds the toy. Players swap roles.

Game 2 - Remote AR Hide-and-Seek



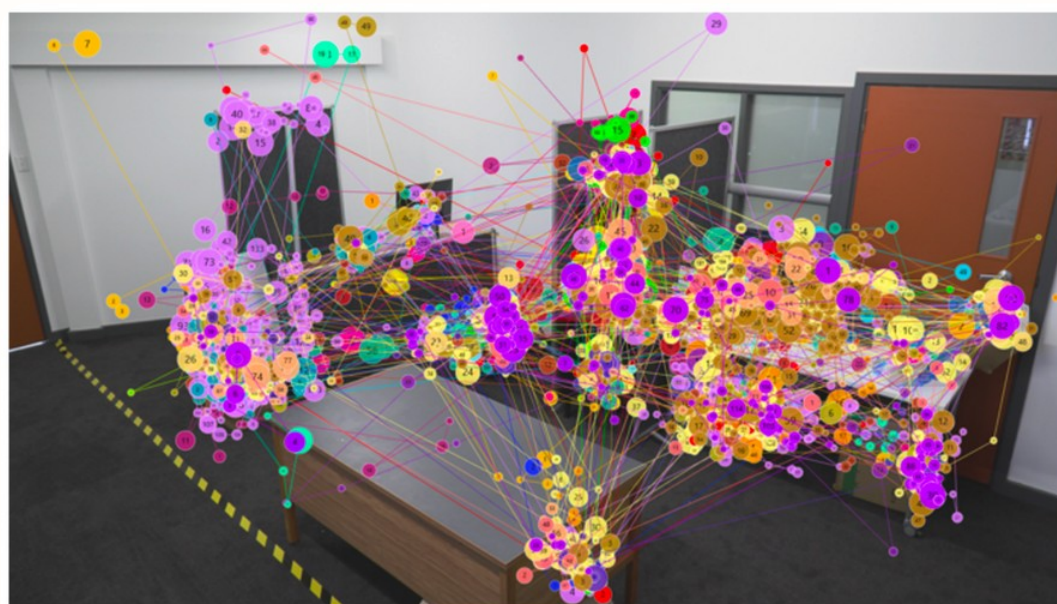
Left: real-world hide-and-seek. Right: remote hide-and-seek in AR.



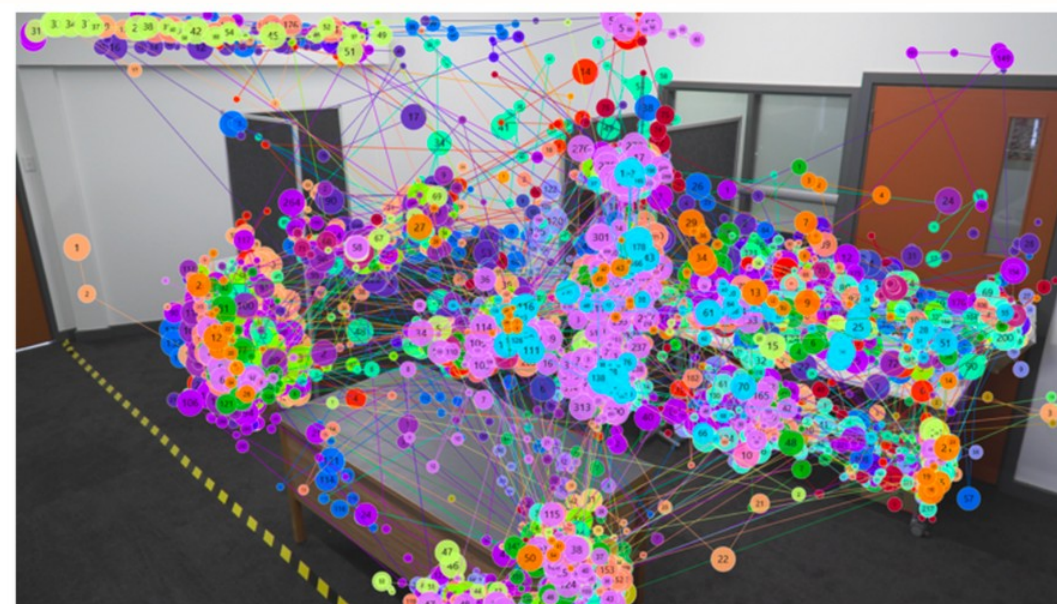
The Hider's view

Game 2 - what we found

AR changed where players hid



RW Seekers' Scan Paths

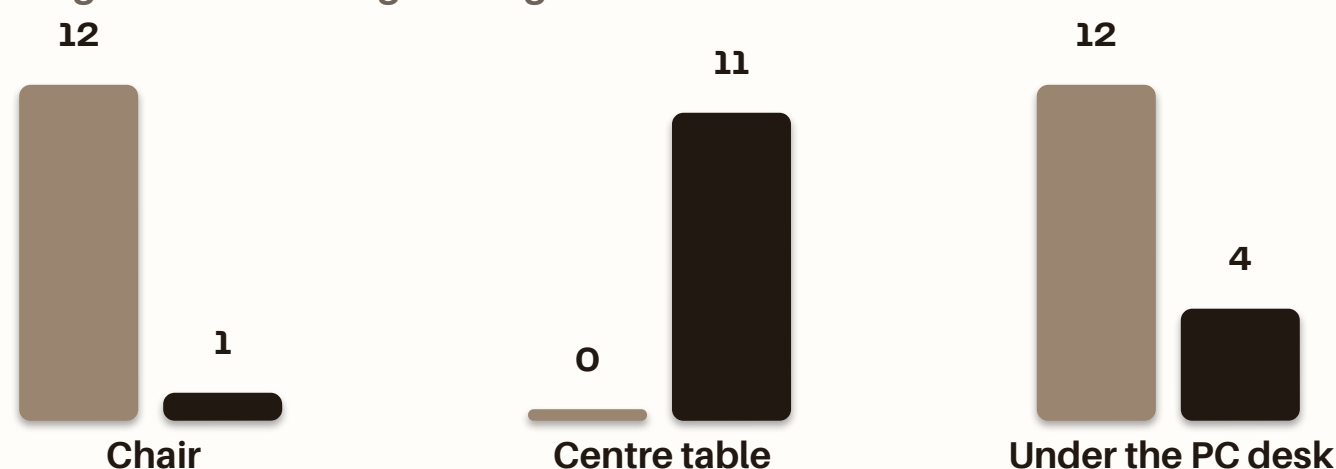


AR Seekers' Scan Paths

Seekers' eye-tracking scan paths — real-world game (left) vs AR game (right)

Where Hiders hid (uses)

■ Real world ■ AR



Hiding took longer

72 s in AR, 52 s in the real world. More places to consider.

Similar find time

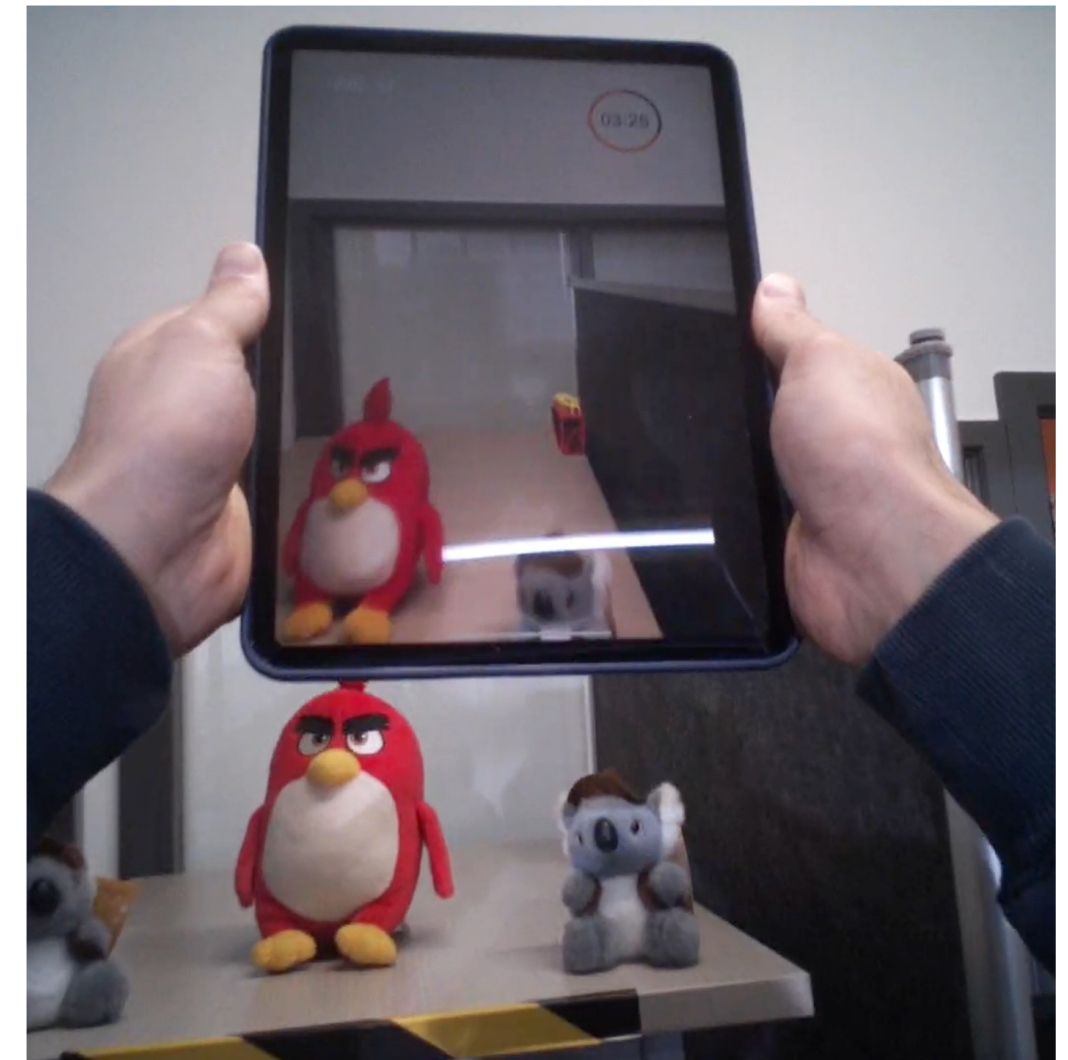
No significant difference. Seekers' gaze spread wider and higher in AR.

32 of 60

players said they lost track of time in AR. "Fun" (hedonic) score: 2.0 in AR vs 0.6 real.

Game 2

Hiding with the real room

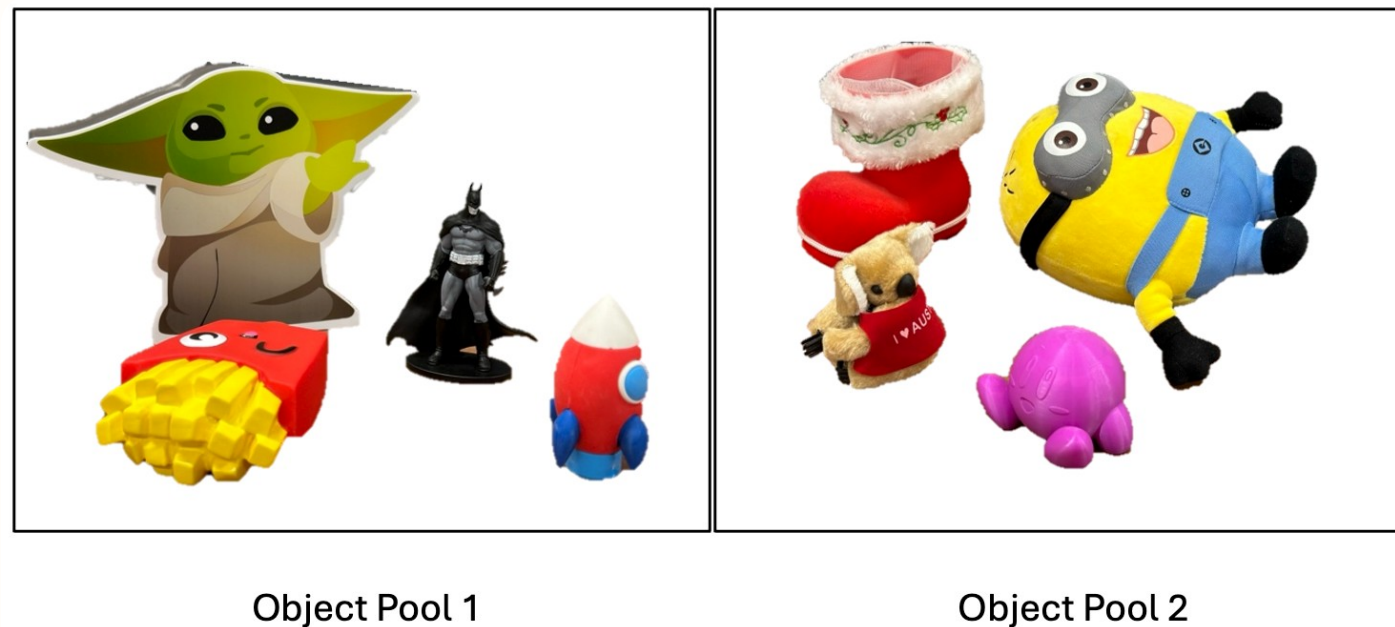


Players hid objects behind real ones and matched colours to the room

Eye-tracker frames · MTI 9(8):79

Game 2 - a surprise

Players hid bright objects more in AR

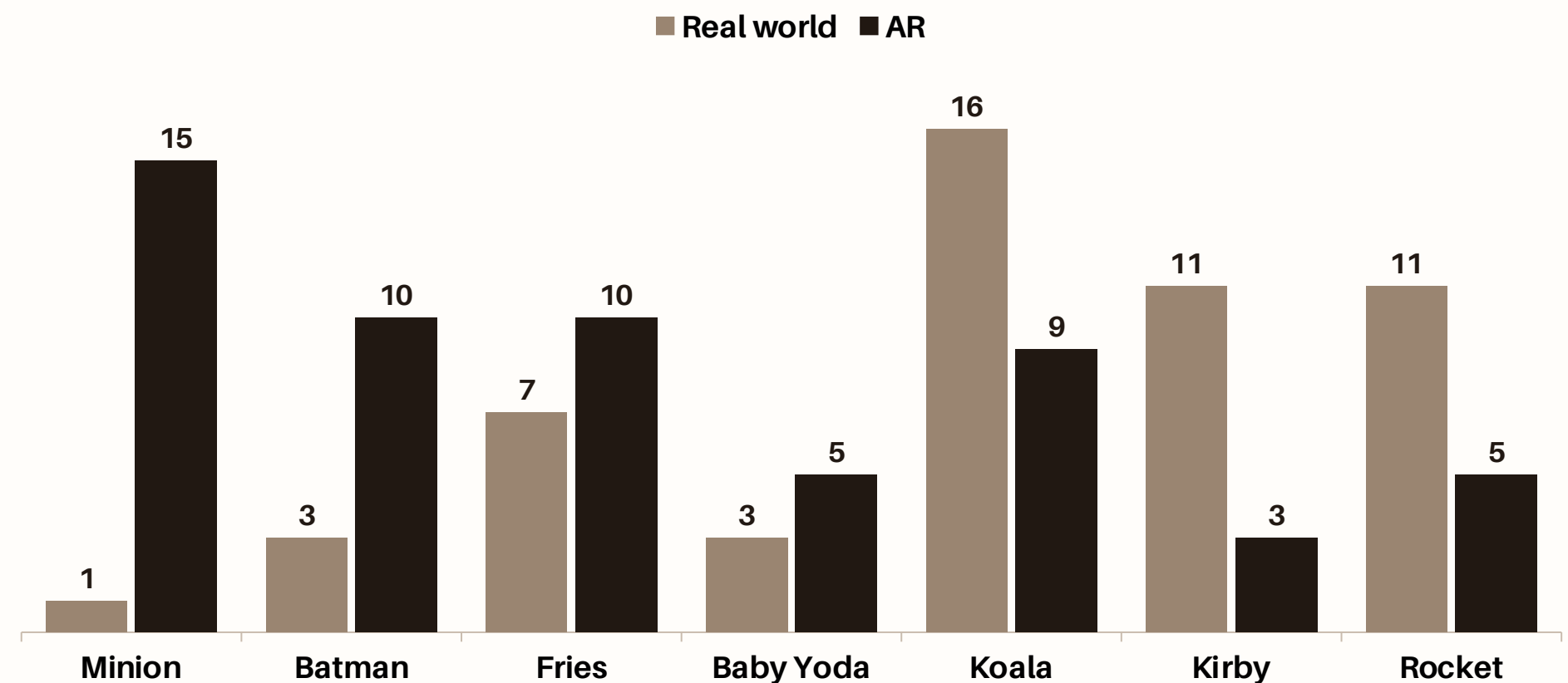


The two object pools Hiders chose from

“The yellow Minion stood out more in AR, so it was easier to place where I wanted.”

Bright, high-contrast objects were picked 1.6× more often in AR, even though the goal was to hide them.

Times each object was chosen to hide



RULE

When controls are hard, players adapt to the controls first. Watch what they choose.

Game 2 - feeling together

Simple cues helped players feel together



The Seeker, seen by the Hider as a moving view-cone

A frustum and voice

No avatars. Hiders watched the frustum and gave hot/cold hints.

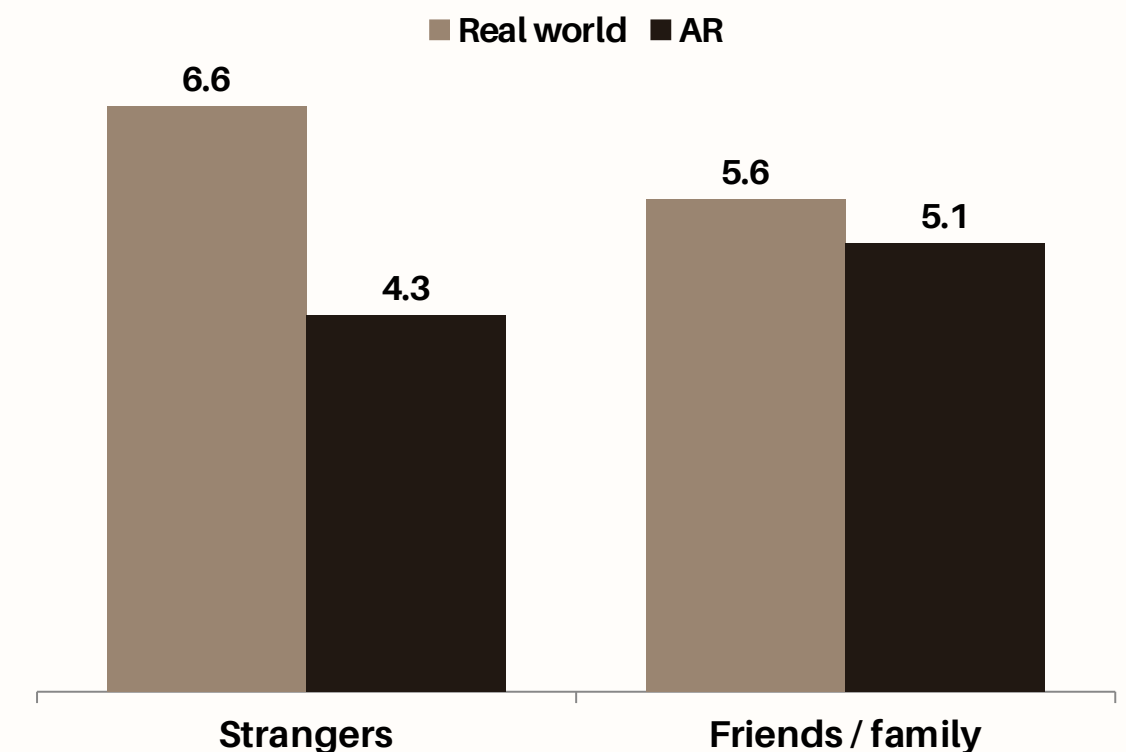
“I live abroad, so it was such a cool way to feel connected to my sister back home.”

“It felt like we were kids again.”

Most didn't need an avatar

Game 3: 16 said the frustum was enough, 10 wanted more.

Verbal hot/cold hints per round



Strangers gave fewer hints in AR.

Friends and family didn't. Give strangers clearer spatial cues.

CHAPTER 3

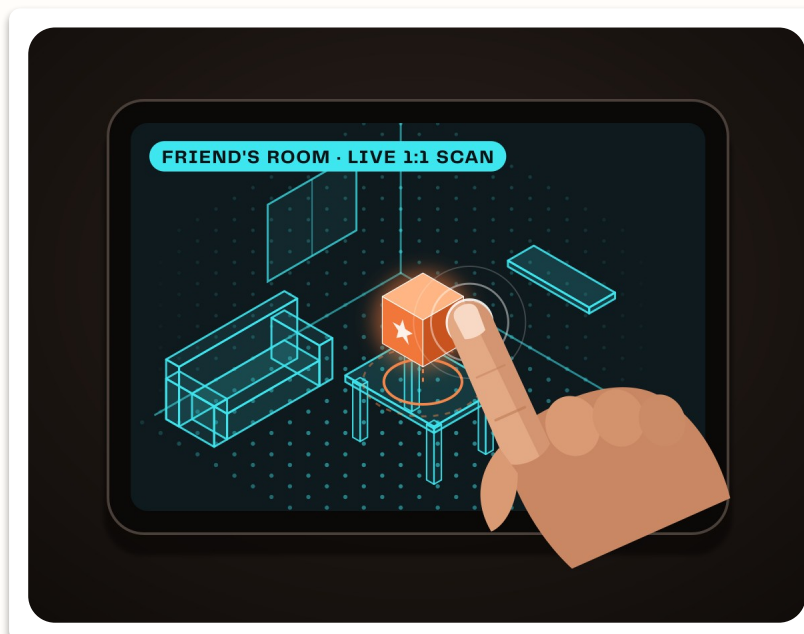
AR becomes the rules

In the third game, we designed mechanics that only work in AR, so the AR can't be switched off.



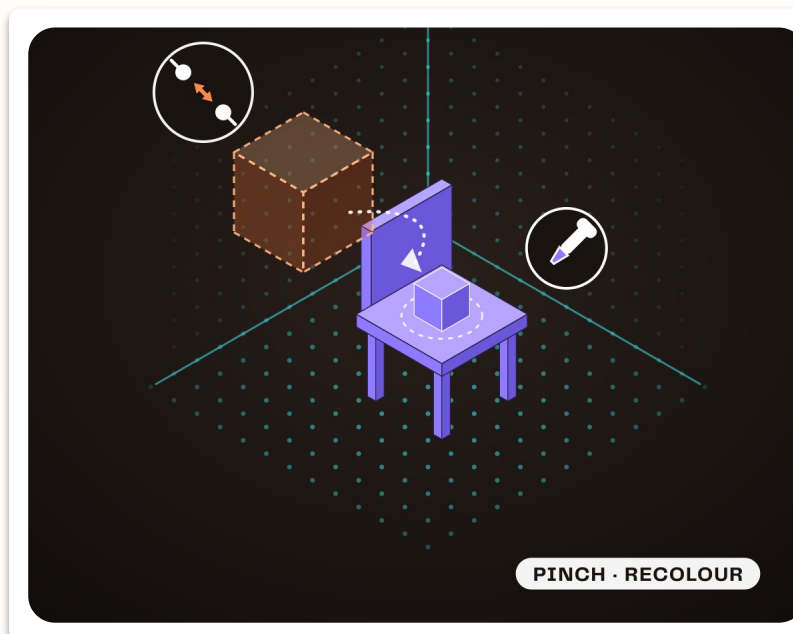
Game 3

Four mechanics that need AR



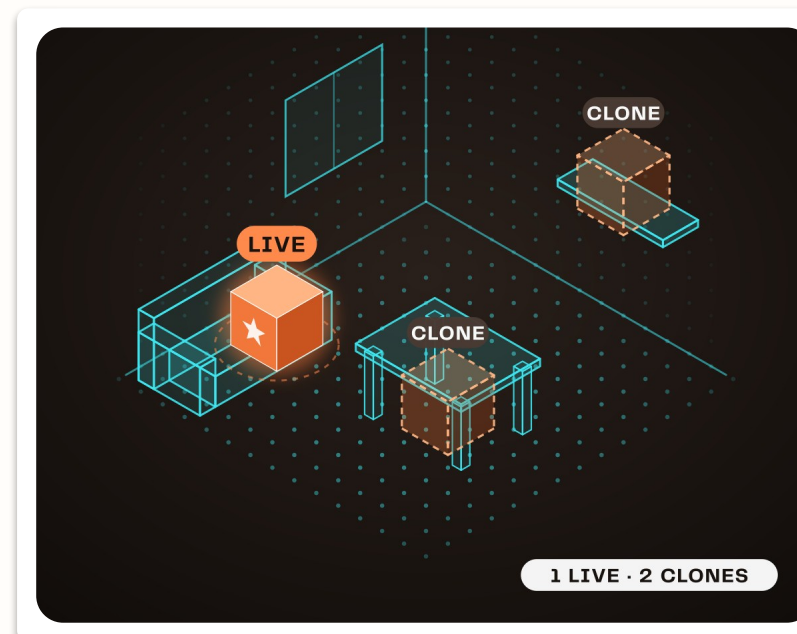
Remote authoring

Hide objects inside a live 1:1 scan of your friend's room.



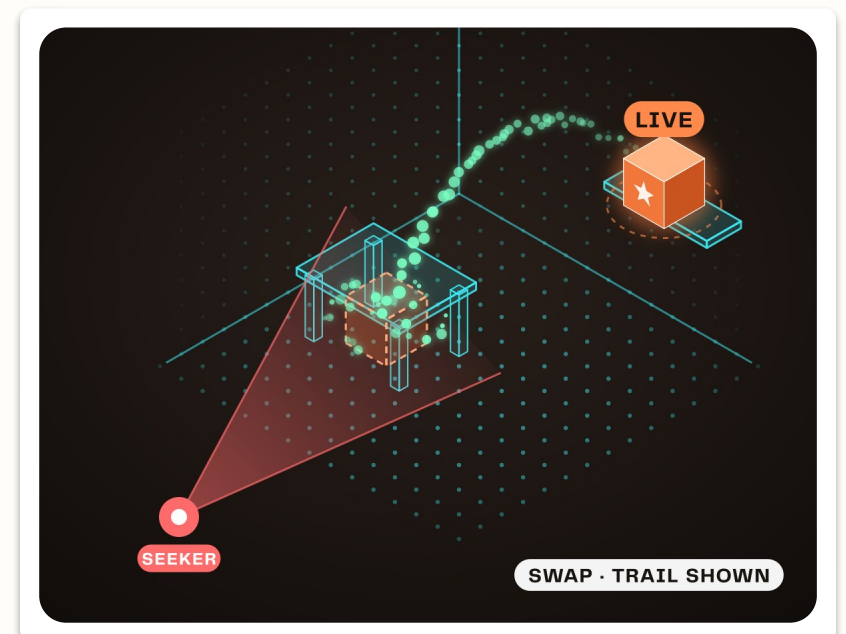
Resize & camouflage

Pinch it tiny, colour-match it to the chair.



Multi-object clones

Hide three copies — only one is live at a time.



Active swapping

Watch their cone get close, then move the live object. A particle trail shows where it was.

“If I see them getting close, I'll just reassign the target to a different spot. It felt a bit sneaky, but that was the point.” — a Hider in Game 3

Game 3 - what we found

Players were split on pacing/timer

ASKED WHICH THEY PREFERRED

9 vs 8

An almost even split.

No single mode won on everything.

FOR A TIMER

“Makes the game more competitive, so it won’t take eternity to finish.”

AGAINST A TIMER

“If I’m playing with friends, I’d rather not have that stress. We just laugh.”

RULE

Let players choose: an optional timer, and swapping as a toggle.

Limitations

What didn't work well

01

Thumbsticks for 3D placement

Usability dropped below the real game. Next time: tap-to-place or auto-snap to surfaces.

02

Overlay needs an empty room

Overlay covers the real world. Some players found it disorienting, which would be a safety issue outdoors.

03

Tablets above your head

Searching the ceiling means holding a tablet up. Four players reported arm fatigue.

04

Advanced mechanics need onboarding

Swapping and resizing needed a proper practice round, not a two-minute demo.

05

Lag and drift

A few players noticed lag and tracking drift, which matters in a game about where things are.

OPEN QUESTION

Does it still work once the novelty wears off?

All our data is from first sessions, indoors, in pairs. Longer, outdoor and 3+ player studies are next.

CHAPTER 4

Building it today

Tools for capturing and sharing real spaces have improved a lot since our studies. Here's what I'd use now.



For developers

How I'd rebuild my games today

01

GAME 1

Splat portals

Realistic Window and Overlay views, still easy to move around in.

Scaniverse · Hyperscape

02

GAME 2

Hide spots from the scan

Hidden objects rest on real shelves and under real tables.

splat-transform

03

LOBBY → PLACE

Instant remote preview

Turns one photo into a 3D view in under a second.

Apple SHARP

04

LOCATION

Portals pinned to places

Both players start from a known landmark in their city.

Niantic VPS 2.0

05

GAME 2 & 3

Your gaze is the cone

Head direction becomes the view-cone. No tablet held up.

AR glasses

06

NEXT

Live rooms, not snapshots

The remote room as it is right now.

GaussianNexus · UIST '25

Summary

Six design rules from these studies

01

If players can switch the AR off, it isn't a mechanic yet.

Make something the game needs that only AR can do.

02

Scale changes presence.

Life-size to feel there. A window to get things done.

03

Break physics on purpose.

Players hid objects on the ceiling and in mid-air, spots the real game doesn't have.

04

A view-cone and voice go a long way.

Try those before building full avatars.

05

Players adapt to your controls first.

Watch what they choose, not just how they score.

06

Let players choose the pacing.

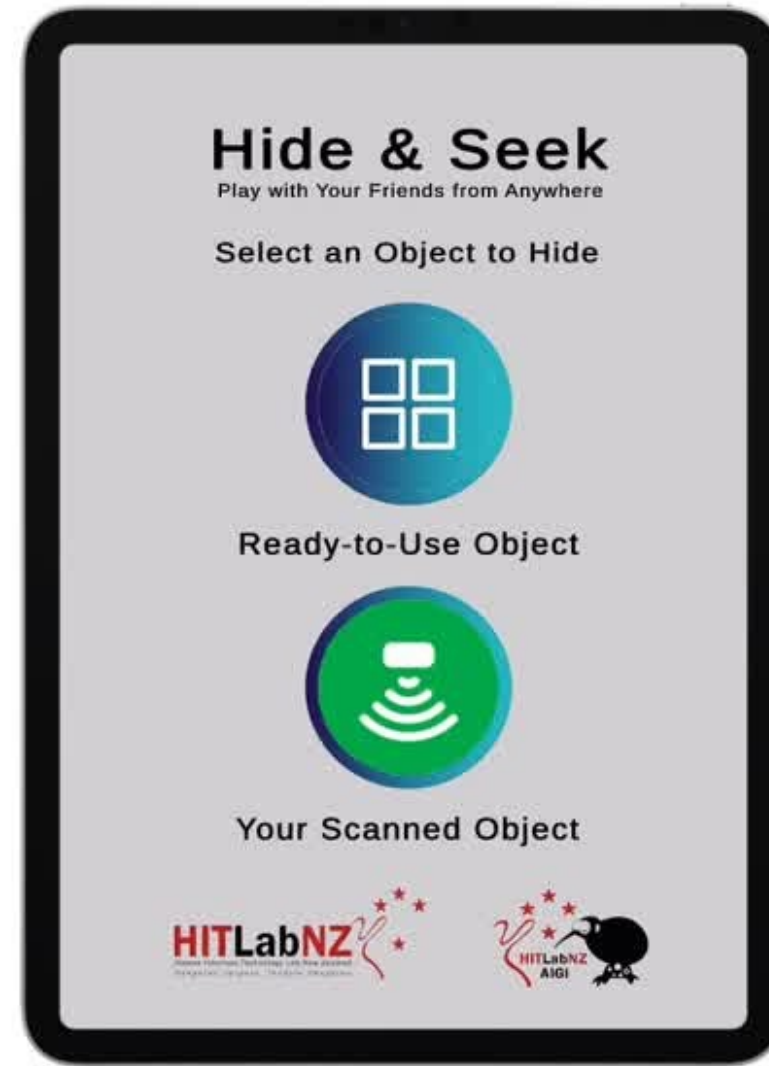
An optional timer, toggles and a practice round.

What's next

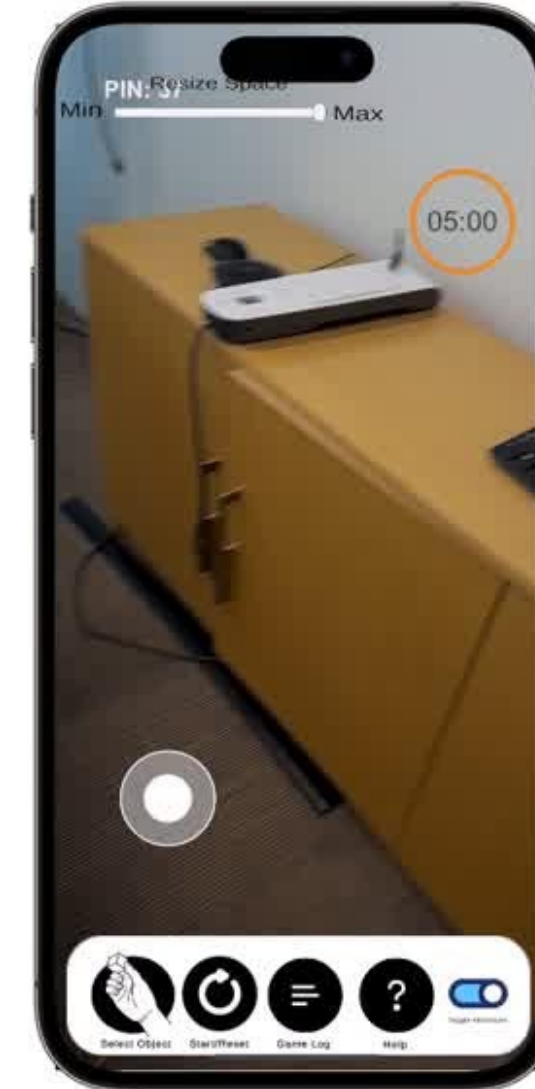
Next: scan a real object, turn it into an AR copy, and hide it in your friend's room.
How do we make that quick and easy?



Hider Scanning Object



Hider Hiding the Object



Seeker Finding the Object

TAKEAWAY

AR becomes a mechanic when the game **stops** **working without it.**

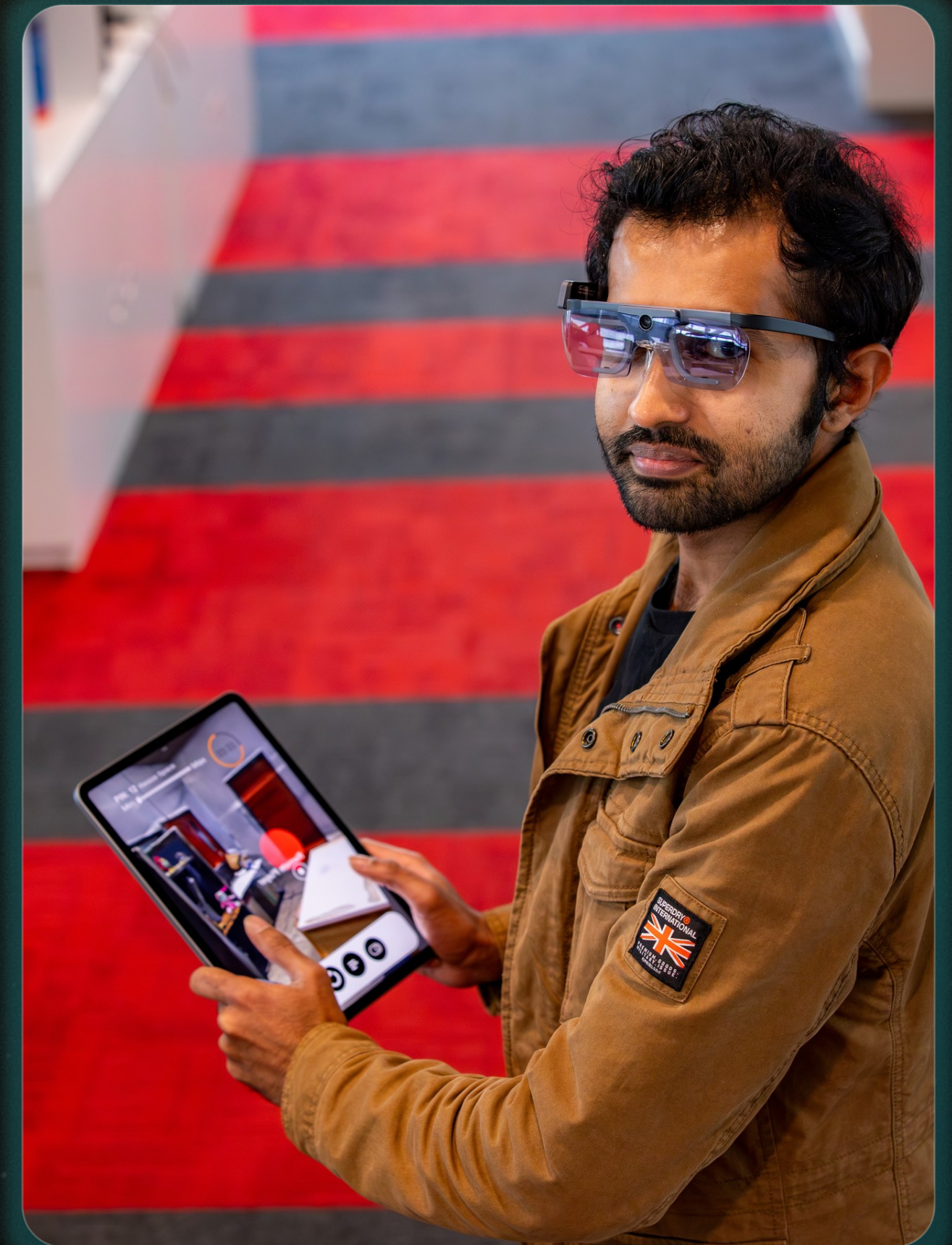
When it changes where players **look** **move** **hide**

Scanning a room now takes a free app.

The harder part is designing good reasons to play in someone else's space, and that's a **game design job.**



yasassri.me/nzgdc
Slides, papers, demos & the six rules



“

**I believe games can transform us.
They can change how we behave,
and the habits we keep,
for the better.**

Yasas Sri Wickramasinghe

Ngā mihi. Questions?

RATE THIS SESSION · NZGDC app · access code **NZgame26**

yasassri.me/nzgdc

Prof. Stephan Lukosch · Prof. Heide Lukosch · James Everett
and heartfelt thanks to everyone who has offered their support...



Scan for everything from this talk

Appendix

References & further reading

MY STUDIES

Wickramasinghe, Lukosch et al. (2025). Representing remote locations with location-based AR game design. Entertainment Computing 53.

Wickramasinghe, Lukosch et al. (2025). Remote multiplayer AR hide-and-seek: spatial decision-making & player experience. Multimodal Technologies and Interaction 9(8), 79.

Wickramasinghe et al. (2025). Augmented hide-and-seek. IEEE VR 2025 Abstracts & Workshops, 1228-1229.

Wickramasinghe et al. (2025). Interactive AR game mechanics for remote multiplayer LBARGs. OzCHI 2025.

Wickramasinghe (2026). PhD thesis, HIT Lab NZ, University of Canterbury.

OTHERS' WORK

Alha, Leorke, Koskinen & Paavilainen (2023). Augmented play. Convergence 29(2), 342-362.

Laato, Inaba & Hamari (2021). Convergence between the real and the augmented. Telematics and Informatics 65, 101716.

Xu, Papangelis et al. (2023). Understanding social interactions in location-based games as hybrid spaces. CHI '23.

Game Informer (Apr 2020) Remote Raid Passes • Sensor Tower (Nov 2020) Pokémon GO 2020 revenue.

Bai et al. (2020). CHI '20 • Jing et al. (2022). PACM HCI • GaussianNexus (2025). UIST '25 • Numan et al. (2024). SpaceBlender. UIST '24.

Niantic Spatial (2026) Scaniverse & VPS 2.0 • Meta (2025) Hyperscape Capture • Apple (2025) SHARP • PlayCanvas splat-transform.

Appendix

Media credits & tools

VIDEOS (click-through, © original creators)

Niantic Spatial — “Scaniverse: The Front Door to Geospatial AI” · youtu.be/BFHAQjLI7rY

Niantic Spatial — “Precise Positioning with VPS 2.0” · youtu.be/LushoeBnQxE

Meta VR — “Meta Horizon Hyperscape Capture (Beta) | Trailer” · youtu.be/zJ4pjwIPiRk

ACM SIGCHI — “GaussianNexus: Room-Scale Real-Time AR/VR Telepresence with Gaussian Splatting” · youtu.be/Zhwz78TgYaw

Peridot (Niantic) — “Peridot Beyond - Multiplayer AR on Snap's Spectacles” · youtu.be/wl10_rJplyc

CNET — “Snap Unveils Specs Smart Glasses at AWE 2026” · youtu.be/fzd-k98ZkLY

Reuters — “Meta unveils Orion AR glasses prototype” · youtu.be/PGuAKyBJ8E8

PCMag — “Watch This Live Demo of Android XR at Google I/O 2026” · youtu.be/HZioERCcZf8

readclub.me — “Building Augmented Reality Applications with Hand Tracking” (clip after the intro, upscaled)

TOOLS & PLATFORMS

Niantic Spatial — Scaniverse, VPS 2.0, Niantic Spatial SDK · nianticspatial.com

Meta — Horizon Hyperscape Capture; Colocation Discovery & Group Sharing · developers.meta.com

Apple — SHARP (ml-sharp) · github.com/apple/ml-sharp

World Labs — Marble; Spark 2.0 renderer · worldlabs.ai · sparkjs.dev

PlayCanvas — SuperSplat, splat-transform, SOG · playcanvas.com

Khronos — KHR_gaussian_splatting (glTF) · khronos.org

Snap — Spectacles Sync Kit, Lens Studio · developers.snap.com

All other photos, videos and figures: the author's own studies (HIT Lab NZ, University of Canterbury).